AV AI VELOCITY ARCHITECT INTELLIGENCE · ISSUE 04

WEEK OF 19–25 SEPTEMBER 2026

After The Yes

A reviewer approved two writes and one ran zero times. A failing write ran twice. A vendor put in writing that recovery is not exactly-once.

22 repos read

3 mechanisms executed

8 papers read

17 models priced

Vijay Zharotia
Healthcare AI architecture

WHY AI VELOCITY

Agents that act inherit obligations.

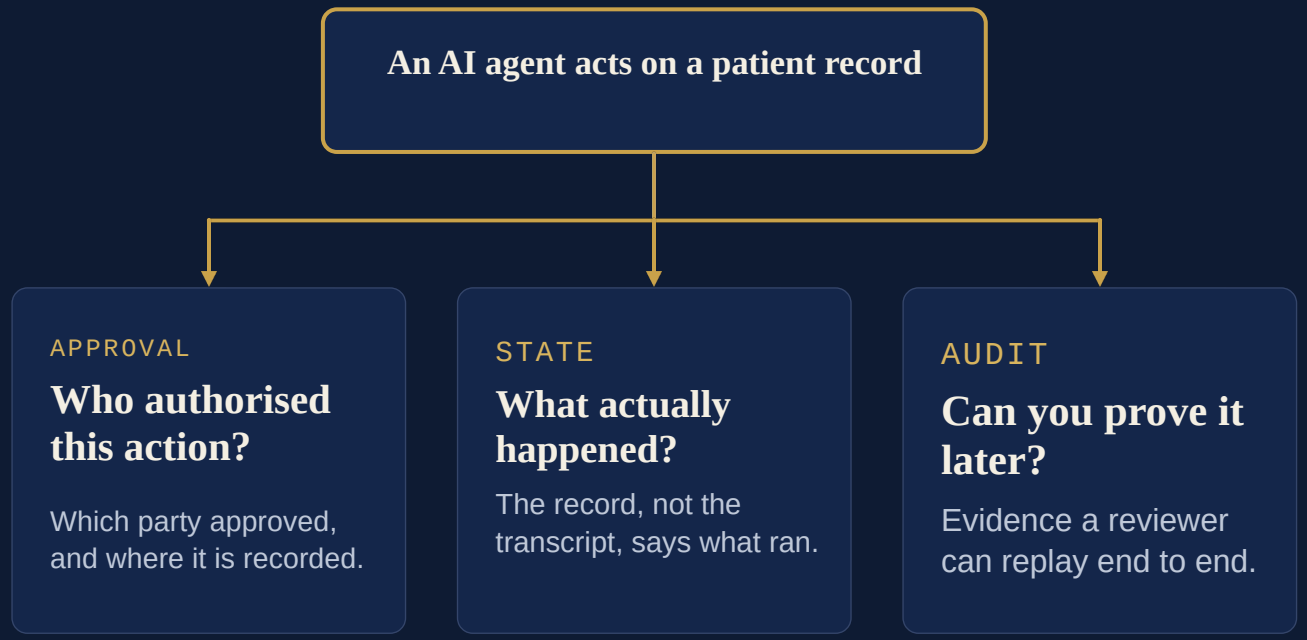
A weekly architecture brief for teams putting AI agents into healthcare workflows. We read the releases, run the mechanism, and say what to change.

FOR ARCHITECTS

The mechanism, the code, the move.

FOR PRODUCT MANAGERS

The decision, the risk, the date.



EVERY CLAIM CARRIES ITS EVIDENCE

EXECUTED

We ran it. The numbers are observed output.

DOCS

Read from documentation or source.

UNVERIFIED

Not confirmed. Vendor numbers carry an asterisk.

THE WEEK IN ONE VIEW

From what the yes covers to how often it runs

CONTEXT

Last week: the yes bound to arguments

Issue 03 asked which arguments a clinician’s approval covers. This week asked a later question: how many times the approved write runs.

PROBLEM

Zero, two, or not promised

An approved write ran zero times. A failing write ran twice. A vendor wrote that recovery does not imply exactly-once execution.

SOLUTION

Join approvals to results

Match every approval to a success result by call id before anything is reported as done, and key every write so it cannot land twice.

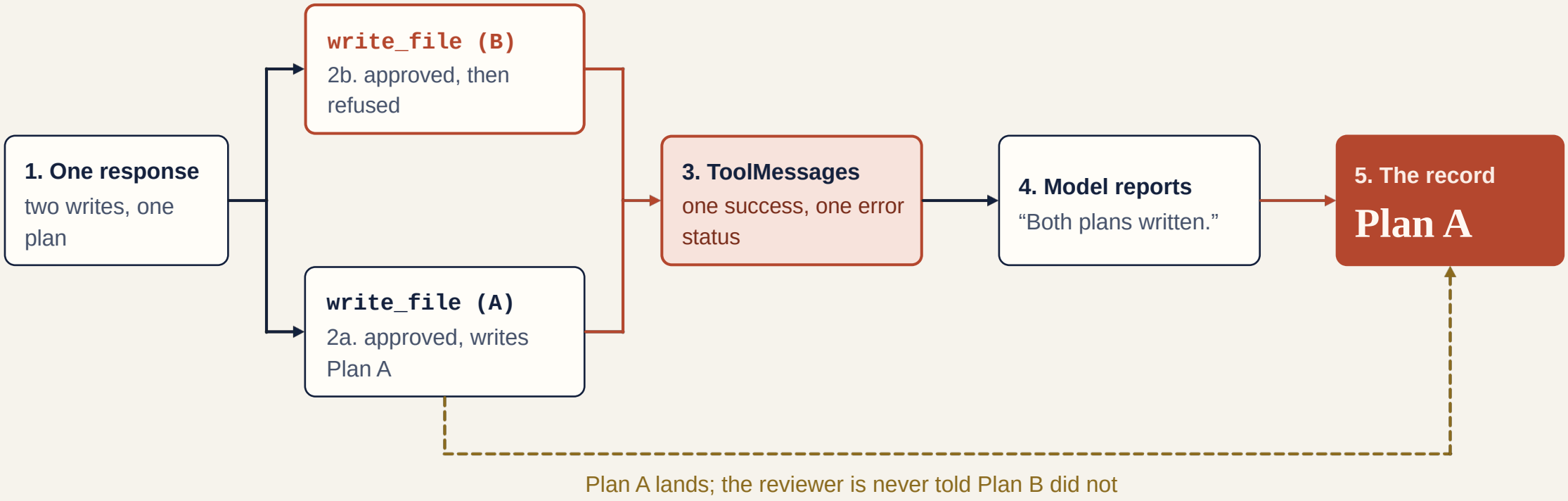
EXECUTION · 04–06
How many times it runs

DATA · 07
Secrets outside the policy

EVIDENCE · 08
Rows nobody counted

COST, DATES · 09–10
Repricing and dockets

The approved write that never ran



EXECUTED MECHANISM · SOLUTION · YOUR HARNESS

Join every approval to its result

STEP 1

Resume after approval

Collect every approved call id.

→

STEP 2

Join to results

Match each to a success ToolMessage by call id.

→

STEP 3

A miss is an incident

Not a retry. Tell the reviewer.

→

STEP 4

Re-run plan evals

First-wins is a different system from last-wins.

2 to 1

writes executed for 2 approvals, 0.7.15 to 0.7.16

0

notices to the reviewer that an approved call was refused

1st wins

the parallel write that wins the record; it used to be the last

> test_da_parallel_writes.py runs both versions in under two minutes. The model said “Both plans written.” on both.

Three places the count went wrong

CONTEXT
how it runs

Reviewer approves → Approval stored → Tool call runs → Result returned → Record written

PROBLEM
where it broke

LANGGRAPH 1.2.12
A dict schema let ‘aprove’ pass as a no
EXECUTED

LLAMA-INDEX
Below 0.14.25 a failing write ran twice
EXECUTED

MS AGENT FW
A child’s pending approval now fails the parent
DOCS

SOLUTION
what to build

Key every write, type every approval

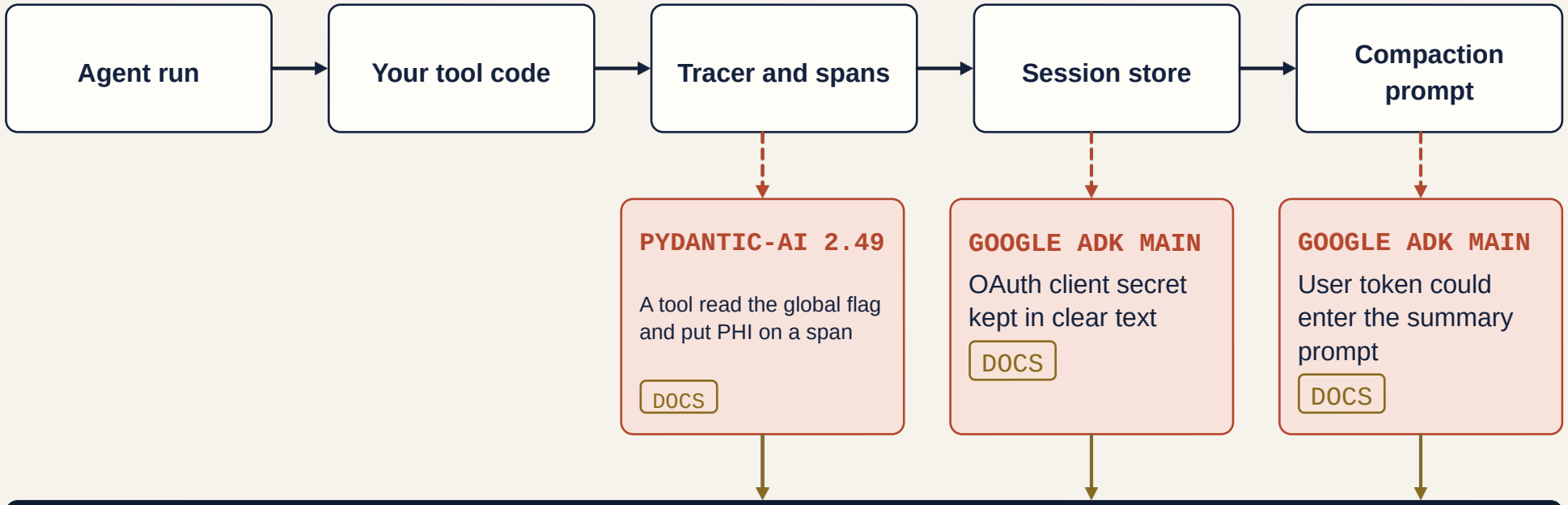
Give every write an idempotency key the tool checks before it commits. Validate approvals with a Pydantic model, never a dict, and make the key cover retries.

Secrets and PHI found a second path

CONTEXT
how it runs

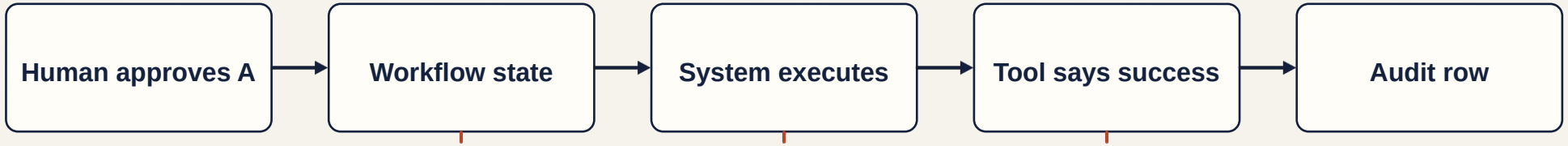
PROBLEM
where it leaked

SOLUTION
what to build



Approve A, execute B, count nothing

CONTEXT
how it runs



PROBLEM
where it broke

2609.21081
7 + 12
Agno and LangGraph server builds ran B

2609.28586
40 to 13
missed effects, before and after metadata

2609.26836
91
tool 'successes' that returned incomplete data

SOLUTION
what to build

Approve the payload, check the result

Hash the full payload at approval and compare it at execution. Write linked effects into the approval row. Validate every tool response for fields, counts and filters.

COST · LIST PRICES READ 25 SEPTEMBER

Opus 5.5 leads both indexes at $260

MODEL	COST PER 10,000 DOCS	GEN / MED*
GPT-6 Luna	$6.50	not read
GPT-6 Sol	$130	48 / 43.5
GPT-5.6 Sol	$260 deprecated	47 / 45.4
Claude Opus 5.5	$260	58 / 60.5
Claude Opus 5	$325 deprecated	not read

Per million tokens in/out: Luna $0.10/$0.50, Sol $2/$10, GPT-5.6 Sol $4/$20 (promotional), Opus 5.5 $4/$20, Opus 5 $5/$25. 4,000 in and 500 out per document, uncached. *Artificial Analysis general and Healthcare & Medical scores as read 25 Sep, not re-verified; Opus 5.5 now reads 61 on medical.

CONTEXT

Opus 5.5 and GPT-6 Sol and Luna all launched on 22 September.

PROBLEM

Sol beat its predecessor on the general index and fell on the medical one. Claude's newer tokenizer makes about 30% more tokens for the same text.

DECISION

Re-tokenise your own documents, price the cached run, and record both scores with the date read.

Two FDA comment windows and a workshop



● REGULATORY ● MARKET AND FUNDING

*Date not confirmed; awards reported 30 Sep to 1 Oct.

Signed evidence found a health-sector home

CHAI

Stewardship of OVERT

An open spec for signed runtime evidence of agent tool calls, taken over from Glacis.

LANGCHAIN

Release gates in healthcare

Abridge’s five-stage gates and clinician annotation queues; Trajectories in LangSmith.

MICROSOFT

Recovery is not exactly-once

Checkpoint-and-recover hosting, plus Foundry egress policies in Audit and Enforced modes.

UK AISI

Verified transcripts

Five benchmarks including HealthBench across six frontier models, in a shared schema.

OPENAI

Rules for outside assessments

Seven principles, and MentalHealthBench with 80-plus clinicians across 19 languages.

QUIET THIS WEEK

HL7, FDA on generative AI, ONC, CMS, NIST, Epic, MCP. Oracle Health launched write-side agents with no published audit contract.

Seven moves, with the reason for each

Move	Why	By
Join each approved call to a success result by call id	Refused writes go unseen	Every resume
Add an idempotency key every write checks before commit	A failing write ran twice	This sprint
Validate approvals with a Pydantic model, never a dict	A typo became a no	Next upgrade
Rotate OAuth client secrets on durable session stores	Stored in clear text	Now
Hash the full payload at approval; compare at execution	Approve A, execute B	Design review
Re-tokenise your own documents before budgeting Opus 5.5	About 30% more tokens	Before budget
File on the FDA robotics and gen-AI dockets	Autonomy vocabulary set here	19 Oct, 24 Nov

AV AI VELOCITY

THIS WEEK’S POSITION

An approval is not a write until you join it to the result.

Long-form edition
Every mechanism, source and date

Lab package
Runs clean, no API key needed

New issue every Friday
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